LEVEL V
CLASS WORK
SINGLE ANSWER QUESTIONS

1. The line which is parallel to x-axis and crosses the curve $y = \sqrt{x}$ at an angle of $\frac{\pi}{4}$ is

A) $y = -\frac{1}{2}$ B) $x = \frac{1}{2}$

C) $y = \frac{1}{4}$ D) $y = \frac{1}{2}$

2. A function $y = f(x)$ is given by $x = \frac{1}{1+t^2}$ & $y = \frac{1}{t(1+t^2)}$ for all $t > 0$ then f is

A) increasing in (0, 3/2) & decreasing in $(3/2, \infty)$

B) increasing in (0, 1)

C) increasing in $(0, \infty)$

D) decreasing in (0, 1)

3. If the normal to the curve y = f(x) at the point (3, 4) makes an angle $\frac{3\pi}{4}$ with the positive x-axis, then $f'(3) = ...$ [IIT 2000]

A) –1 B) $-\frac{3}{4}$ C) $\frac{4}{3}$ D) 1

4. Let $f(x) = \begin{cases} |x|, & \text{for } 0 < |x| \le 2 \\ 1, & \text{for } x = 0 \end{cases}$ then at x = 0, f has [IIT 2000]

A) a local maximum B) no local maximum

C) a local minimum D) no extremum

5. For all $x \in (0,1)$ [IIT 2001]

A) $e^x < 1 + x$ B) $\log_e(1 + x) < x$

C) $\sin x > x$ D) $\log_e x > x$

6. If $f(x) = xe^{x(1-x)}$, then f(x) is [IIT 2001]

A) increasing on [–1/2, 1]

B) decreasing on R

C) increasing on R

D) decreasing on [–1/2, 1]

7. $f(x) = 2.e^{x^2-4x}$ decreases in

A) $(2, \infty)$ B) $(2, -\infty)$ C) $(-\infty, 2)$ D) R

8. The values 'a' for which the function $f(x) = (a + 2)x^3 - 3a\,x^2 + 9a\,x - 1$ decreases for all real values of x,is

A) a < -2 B) a > -2

C) a < -3 D) -3 < a < -2

9. Maximum value of f(x) = x + sin 2x, $x \in [0, 2\pi]$ is

A) π B) 2π C) 3π D) $\pi/2$

10. Minimum value of $f(x) = x^2 \log x$, $x \in [1, e]$ is

A) 0 B) 1 C) 2 D) 3

11. Consider the curve $y = c\, e^{\frac{x}{a}}$. The equation of normal to the curve where the curve cut y-axis is

A) $cx + ay = c^2$ B) $cy + ax = c^2$

C) x + y = 2 D) x + y = 4

12. Which of the following function does not satisfies the condition for requirement of Lagrange's Mean Value Theorem (LMVT]

A) $f(x) = \begin{cases} \frac{\sin 3x}{x}, & 0 < x \le 5 \\ 3, & x = 0 \end{cases}, x \in [0, 5]$

B) $f(x) = \tan\frac{\pi x}{2}, x \in \left[-\frac{1}{2}, \frac{1}{2}\right]$

C) $f(x) = \begin{cases} x^3 \sin\frac{1}{x}, & 0 < x \le 1 \\ 0, & x = 0 \end{cases}, x \in [0, 1]$

D) $f(x) = \begin{cases} x^2 + \sin x, & x < 0 \\ x^3 - 5x, & x \ge 0 \end{cases}, x \in [-2, 3]$

13. Rolle's theorem is not applicable for the function f(x) = |x| in the interval [–1, 1] because

A) $f'(1)$ does not exist

B) $f'(-1)$ does not exist

C) f(x) is discontinuous at $x=0$

D) $f'(0)$ does not exist

14. The greatest value of the function

$f(x)=\dfrac{\sin 2x}{\sin\left(x+\dfrac{\pi}{4}\right)}$ **on the interval** $\left[0,\dfrac{\pi}{2}\right]$

is

A) $\dfrac{1}{\sqrt{2}}$ B) $\sqrt{2}$ C) 1 D) $-\sqrt{2}$

15. The range of y = (arc cosx) (arc sinx) is

A) $\left[-\dfrac{\pi}{2},\dfrac{\pi}{2}\right]$ B) $\left(-\dfrac{\pi}{2},\dfrac{\pi}{2}\right)$

C) $\left[\dfrac{-\pi^2}{2},\dfrac{\pi^2}{16}\right]$ D) $\left[0,\dfrac{\pi^2}{16}\right]$

16. Rolle's theorem holds for the function

$x^3+bx^2+cx,\ 1\le x\le 2$ **at the point x = 4/3, the values of b and c are**

A) $b=8, c=-5$ B) $b=-5, c=8$

C) $b=5, c=-8$ D) $b=-5, c=-8$

17. If a, b, c, d are real numbers such that

$\dfrac{3a+2b}{c+d}+\dfrac{3}{2}=0$, **Then the equation**

$ax^3+bx^2+cx+d=0$ has.

A) at least one root in $[-2, 0]$

B) at least one root in $[0, 2]$

C) at least two roots in $[-2, 2]$

D) No root in $[-2, 2]$

18. If $y = a\ln|x| + bx^2 + x$ has its local extremum values at $x = -1$ and $x = 2$, then

A) $a=2, b=-1$ B) $a=2, b=-\dfrac{1}{2}$

C) $a=-2, b=\dfrac{1}{2}$ D) $a=2, b=1/2$

19. The equations of the tangents to the curve $y = x^4$ from the point (2, 0) other than x-axis, is

A) $y=0$ B) $y-1=5(x-1)$

C) $y-\dfrac{4096}{81}=\dfrac{2048}{27}\times\left(x-\dfrac{8}{3}\right)$

D) $y-\dfrac{32}{243}=\dfrac{80}{81}\times\left(x-\dfrac{2}{3}\right)$

20. The greatest value of the expression

P(x) $=(1-x)^5(1+x)(1+2x)^2$

A) 3 B) 1 C) 4 D) 6

21. Which one of the following curves cut the parabola $y^2 = 4ax$ at right angles?

[IIT 1994]

A) $x^2+y^2=a^2$ B) $y=e^{-x/2a}$

C) $y=ax$ D) $x^2=4ay$

22. The function defined by $f(x) = (x + 2)e^{-x}$ is

[IIT 1995]

A) decreasing for all x

B) decreasing in $(-\infty,-1)$ and increasing in $(-1,\infty)$

C) increasing for all x

D) decreasing in $(-1,\infty)$ and increasing in $(-\infty,-1)$

23. On the interval [0, 1] the function $x^{25}(1-x)^{75}$ takes its maximum value at the point [IIT 1995]

A) 0 **B)** $\dfrac{1}{4}$ C) $\dfrac{1}{2}$ D) $\dfrac{1}{3}$

24. If $f(x)=\dfrac{x}{\sin x}$ and $g(x)=\dfrac{x}{\tan x}$, **where** $0<x\le 1$, **then in this interval**

A) both f(x) and g(x) are increasing functions

B) both f(x) and g(x) are decreasing functions

C) f(x) is an increasing function

D) g(x) is an increasing fucntion

25. The function $f(x) = \sin^4 x + \cos^4 x$ increases if

[IIT 1999]

A) $0<x<\dfrac{\pi}{8}$ B) $\dfrac{\pi}{4}<x<\dfrac{3\pi}{8}$

C) $\dfrac{3\pi}{8}<x<\dfrac{5\pi}{8}$ D) $\dfrac{5\pi}{8}<x<\dfrac{3\pi}{4}$

26. Let $f(x)=\int e^x(x-1)(x-2)dx$. **Then f decreases in the interval [IIT 2000]**

A) $(-\infty,-2)$ B) $(-2,-1)$

C) $(1, 2)$ D) $(2, \infty)$

27. **If $2a + 3b + 6c = 0$, then the equation $ax^2 + bx + c = 0$ has at least one real root in**

A) (0, 1) B) (0, 1/2) C) (1/4, 1/2) D) (-1, 1)

28. **Let f be differentiable for all x. If $f(1) = -2$ and $f'(x) \geq 2$ for all $x \in [1, 6]$. Then**

A) $f(6) = 8$ B) $f(6) \geq 8$

C) $f(6) \leq 8$ D) $f(6) < 8$

29. **If an interval (a, b) contains k roots of a real polynomial P(x) then it, contains**

A) at least (k - 1) roots of $P'(x) = 0$

B) at most (k - 1) roots of $P'(x) = 0$

C) at most k roots of $P'(x) = 0$

D) atleast k roots of $P'(x) = 0$

30. **Let α, β $(\alpha < \beta)$ be two real roots of the equation $ax^2 + bx + c = 0$. Then $\frac{-b}{2a}$ lies in**

A) (β, α) B) $(-\alpha, \alpha)$

C) $(-\beta, \beta)$ D) (α, β)

31. **The triangle formed by the tangent to the curve $f(x) = x^2 + bx - b$ at the point (1, 1) and the coordinate axes, lies in the first quadrant. If its area is 2, then the value of b is [IIT 2001]**

A) –1 B) 3 C) –3 D) 1

32. **Let $f(x) = (1 + b^2)x^2 + 2bx + 1$ and let m(b) be the minimum value of f(x). As b varies, the range of m(b) is**

A) [0, 1] B) [0, 1/2] C) [1/2, 1] D) (0, 1]

33. **If a variable tangent to the curve $x^2y = c^3$ makes intercepts a, b on x and y axis respectively, then the value of a^2b is**

A) $27c^3$ B) $\frac{4}{27}c^3$ C) $\frac{27}{4}c^3$ D) $\frac{4}{9}c^3$

34. **If $f(x) = x^3 + 7x - 1$ then $f(x)$ has a zero between x = 0 and x = 1. The theorem which best describes this, is**

A) Squeeze play theorem

B) Mean value theorem

C) Maximum-Minimum value theorem

D) Intermediate value theorem

35.

$$f(x) = \begin{cases} x\sin\frac{\pi}{x} & \text{for } x > 0 \\ 0 & \text{for } x = 0 \end{cases}$$

then the number of points in (0, 1) where the derivative $f'(x)$ vanishes, is

A) 0 B) 1 C) 2 D) infinite

36. **Suppose that $f(0) = -3$ and $f'(x) \leq 5$ for all values of x. Then the largest value which f (2) can attain is**

A) 7 B) – 7 C) 13 D) 8

37. **Equation of the line through the point (1/2, 2) and tangent to the parabola $y = \frac{-x^2}{2} + 2$ and secant to the curve $y = \sqrt{4 - x^2}$ is**

A) $2x + 2y - 5 = 0$ B) $2x + 2y - 3 = 0$

C) $y - 2 = 0$ D) $2x - 1 = 0$

38. **A curve is represented by the equations, $x = \sec^2 t$ and $y = \cot t$ where t is a parameter. If the tangent at the point P on the curve where $t = \pi/4$ meets the curve again at the point Q then $|PQ|$ is equal to**

A) $\frac{5\sqrt{3}}{2}$ B) $\frac{5\sqrt{5}}{2}$ C) $\frac{2\sqrt{5}}{3}$ D) $\frac{3\sqrt{5}}{2}$

39. **For all $a, b \in R$ the function $f(x) = 3x^4 - 4x^3 + 6x^2 + ax + b$ has**

A) no extremum

B) exactly one extremum

C) exactly two extremum

D) three extremum

40. **The set of values of p for which the equation $|\ln x| - px = 0$ possess three distinct roots is**

A) $\left(0, \frac{1}{e}\right)$ B) (0, 1) C) (1,e) D) (0,e)

41. **A point is moving along the curve $y^3 = 27x$. Then the interval of values of x in which the ordinate changes faster than abscissa is**

A) $x \in (-1,\ 1)$ B) $x \in (-1,\ 1) - \{0\}$

C) $x \in [-1,\ 1] - \{0\}$ D) $x \in (-1,1]$

42. An edge of a variable cube is increasing at the rate of 3 cm per second. How fast is the volume of the cube increasing when the edge is 10 cm long ?

A) $100 cm^3/sec$ B) $900 cm^3/sec$

C) $800 cm^3/sec$ D) $1200 cm^3/sec$

43. Let $f(x) = \begin{cases} -x^2 & \text{for } x < 0 \\ x^2 + 8 & \text{for } x \ge 0 \end{cases}$. Then x intercept of the line that is tangent to the graph of $f(x)$ is

A) zero B) – 1 C) –2 D) – 4

44. There are 50 apple trees in an orchard. Each tree produces 800 apples. For each additional tree planted in the orchard, the output per additional tree drops by 10 apples. Number of trees that should be added to the existing orchard for maximising the output of the trees, is

A) 5 B) 10 C) 15 D) 20

45. The ordinate of all points on the curve $y = \dfrac{1}{2\sin^2 x + 3\cos^2 x}$ where the tangent is horizontal, is

A) always equal to 1/2

B) always equal to 1/3

C) 1/2 or 1/3 according as *n* is an even or an odd integer.

D) 1/2 or 1/3 according as *n* is an odd or an even integer.

MULTI ANSWER QUESTIONS

46. If the line ax + by + c = 0 is a normal to the rectangular hyperbola xy = 1, then

A) a > 0, b > 0 B) a > 0, b < 0

C) a < 0, b > 0 D) a < 0, b < 0

47. Let

$f(x) = 2\sin^3 x - 3\sin^2 x + 12\sin x + 5,\ 0 \le x \le \pi/2$.

Then f(x) is

A) decreasing in $[0,\ \pi/2]$

B) increasing in $[0,\ \pi/2]$

C) increasing in $[0,\ \pi/4]$ and decreasing in $[\pi/4,\ \pi/2]$

D) none of these

48. Let $f(x) = \dfrac{x^2+1}{[x]},\ 1 \le x \le 3.9$. $[.]$ denotes the greatest integer function. Then

A) f(x) is monotonically decreasing in [1, 3.9]

B) f(x) is monotonically increasing in [1, 3.9]

C) the greatest value of f(x) is $\dfrac{1}{3} \times 16.21$

D) the least value of f(x) is 2.

49. Let $h(x) = f(x) - (f(x))^2 + (f(x))^3$ for every real number x. Then [IIT 1998]

A) h is increasing whenever f is increasing

B) h is increasing whenever f is decreasing

C) h is decreasing whenever f is decreasing

D) nothing can be said in general

50. Let the parabolas $y = x^2 + ax + b$ and $y = x(c - x)$ touch each other at a point (1, 0). Then

A) a = – 3 B) b = 1

C) c = 2 D) b + c = 3

51. A point on the ellipse $4x^2 + 9y^2 = 36$ where the tangent is equally inclined to the axes is

A) $\left(\dfrac{9}{\sqrt{13}}, \dfrac{4}{\sqrt{13}}\right)$ B) $\left(-\dfrac{9}{\sqrt{13}}, \dfrac{4}{\sqrt{13}}\right)$

C) $\left(\dfrac{9}{\sqrt{13}}, -\dfrac{4}{\sqrt{13}}\right)$ D) $\left(\dfrac{4}{\sqrt{13}}, -\dfrac{9}{\sqrt{13}}\right)$

52. If $f(x) = \begin{cases} 3x^2 + 12x - 1, & -1 \le x \le 2 \\ 37 - x, & 2 < x \le 3 \end{cases}$ **then [IIT-1993]**

A) f(x) is increasing on [–1, 2]

B) f(x) is continous on [–1, 3]

C) $f'(2)$ doesn't exist

D) f(x) has the maximum value at x = 2

53. Let $f(x) = x^3 + ax^2 + bx + 5\sin^2 x$ be an increasing function in the set of real numbers R. Then a and b satisfy the condition

A) $a^2 - 3b - 15 \le 0$ B) $a^2 - 3b + 15 \ge 0$

C) $a^2 - 3b + 15 \le 0$ D) a > 0 and b > 0

54. Let $f(x) = (x-1)^4.(x-2)^n,\ n \in N$. Then f(x) has

A) a local maximum at $x = 1$ if n is odd
B) a local maximum at $x = 1$ if n is even
C) a local minimum at $x = 2$ if n is even
D) a local maximum at $x = 2$ if n is odd

55. Let $f(x)=\begin{cases} x^3+x^2-10x & -1\le x<0 \\ \sin x & 0\le x<\frac{\pi}{2} \\ 1+\cos x & \frac{\pi}{2}\le x\le \pi \end{cases}$

then f(x) has

A) local maxima at $x=\frac{\pi}{2}$
B) local minima at $x=\frac{\pi}{2}$
C) absolute maxima at $x = 0$
D) absolute maxima at $x = -1$

56. **If** $f(x)=\sin x, -\pi/2\le x\le \pi/2$ **, then**
A) f(x) is increasing in the interval $[-\pi/2,\pi/2]$
B) f{f(x)} is increasing in the interval $[-\pi/2,\pi/2]$
C) f{f(x)} is decreasing in $[-\pi/2,0]$ and increasing in $[0,\pi/2]$
D) f{f(x)} is invertible in $[-\pi/2,\pi/2]$

57. **If** $f(x)=x^4-4x-1$, **then**
A) f (x) has exactly one positive real root
B) f (x) has exactly one negative real root
C) f (x) has exactly two real roots
D) minimum value of f (x) = -3

58. **The extremum values of the function** $f(x)=\frac{1}{\sin x+4}-\frac{1}{\cos x-4}$, **where** $x\in R$ **is**

A) $\frac{4}{8-\sqrt{2}}$ B) $\frac{2\sqrt{2}}{8-\sqrt{2}}$ C) $\frac{2\sqrt{2}}{4\sqrt{2}+1}$ D) $\frac{4\sqrt{2}}{8+\sqrt{2}}$

59. **If** $f''(x)<0\,\forall x\in R$ **and** $g(x) = f(x^2 - 2) + f(6 - x^2)$ **then**
A) g(x) is an increasing in [0, 2]
B) g(x) is an increasing in [2, ∞)
C) g(x) has a local minima at $x = -2$
D) g(x) has a local maxima at $x = 2$

60. **The function**

$$f(x)=\int_{-1}^{x} t(e^t-1)(t-1)(t-2)^3(t-3)^5\,dt$$

has a local minimum at x =
A) 0 B) 1 C) 2 D) 3

61. **The function** $y=\frac{2x-1}{x-2}$ $(x\neq 2)$
A) is its own inverse
B) decreases for all values of x
C) has a graph entirely above x-axis
D) is bound for all x.

62. **The function** $\frac{\sin(x+a)}{\sin(x+b)}$ **has no maxima or minima if**
A) $b-a=n\pi$, $n\in I$
B) $b-a=(2n+1)\pi$, $n\in I$
C) $b-a=2n\pi$, $n\in I$
D) $b-a=(2n+1)\frac{\pi}{2}$, $n\in I$

MATRIX MATCH QUESTIONS

63. **Let** $f(x) = (2^x - 1)(2^x - 2)$ **and** $g(x) = 2\sin x + \cos 2x$ **in** $[0, \pi]$

Column-I	**Column-II**
A) f increases on	P) $(\log_2(3/2), \infty)$
B) f decreases on	Q) $(-\infty, \log_2(3/2))$
C) g decreases on	R) $\left(0, \frac{\pi}{6}\right)$
D) g increases on	S) $\left(\frac{5\pi}{6}, \pi\right)$

64. **Column - I gives functions which satisfy conditions of CMVT an specified interval and Column - II gives value of 'C' for which LMVT is satisfied**

Column - I	**Column - II**
A) $f(x)=x(x-2)$ **in** $[1,2]$	P) $\frac{5}{6}$
B) $f(x)=x(2-x)$ in $[0,1]$	Q) $\frac{1}{3}$
C) $f(x)=x^3-2x^2-x+3$ in $[0,1]$	R) 3

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D) $f(x)=(x-1)(x-2)(x-3)$ in $[1,4]$ S) $\frac{7}{6}$

T) A rational number

65. **Match the following**

Column - I **Column - II**

A) $\frac{(x-1)(x-2)}{x-3}$ P) Decreases when $x \in (1-\sqrt{2}, 1+\sqrt{2})$

B) $\frac{(x-2)(x-3)}{x-3}$ Q) Decreases when $x \in (3-\sqrt{2}, 3+\sqrt{2})$

C) $\frac{(x-1)(x-3)}{x-2}$ R) Decreases when $x \in R$

D) $\frac{x-2}{(x-1)(x-3)}$ S) Increases when $x \in R$

T) is monotonic function

66. **Match the following**

Column - I **Column - II**

A) $f(x)=\sin x+\cos x+2x$ strictly increases on P) $(3,\infty)$

B) $f(x)=\frac{x^2+x+1}{x^2-x+1}$ strictly increases on Q) $(1,\infty)$

C) $f(x)=\frac{x^2+x+1}{x^2-x+1}$ strictly decreases on Q) $(1,\infty)$

D) $f(x)=\frac{x^3}{x^4+27}$ strictly decreases on R) $(-\infty,-1)$

S) $(-1,1)$

T) $(-\infty,\infty)$

67. **Match the following**

Column - I **Column - II**

A) $f(x)=\frac{x}{1+x\tan x}$ has maximum value when $x=$ P) Any real value in the domain of definition of function

B) $f(x)=\left(\frac{x}{1+x\tan x}\right)^{-1}$ has minimum value when $x=$ Q) $\frac{\pi}{4}$

C) $f(x)=\left(\frac{x}{1+x\cot x}\right)$ is monotonically increasing when $x=$ R) $\cos x$

D) $f(x)=\left(\frac{x}{1+x\cot x}\right)^{-1}$ has minimum value in $\left(0,\frac{\pi}{4}\right)$ then $x=$ S) $\sin x$

T) The x-coordinate of point of intersection of $y=x$ and $y=\cos x$

68. **Match the max / min value of function in Col - I with corresponding values in Col - II**

Column - I **Column - II**

A) Greatest value of $f(x)=\frac{x}{4+x+x^2}$ on $[0,\infty)$ is P) $\frac{18}{e}$

B) Maximum value of $\frac{\ln x}{x}$ in $[2,\infty)$ is Q) $\frac{1}{e}$

C) Let $x>0, y>0$ & $xy=1$ then minimum value of $\frac{3}{e^3}x+27ey$ is R) e

D) Perimeter of a sector is $4e$. S) $\frac{1}{5}$

The area of sector is maximum when its radius is T) An irrational number

ASSERTION & REASON QUESTIONS

INSTRUCTIONS: In the following questions an Assertion (A) is given followed by a Reason (R). Mark your responses from the following options.

(A) Both Assertion and Reason are true and Reason is the correct explanation of 'Assertion'

(B) Both Assertion and Reason are true and Reason is not the correct explanation of 'Assertion'

(C) Assertion is true but Reason is false

(D) Assertion is false but Reason is true

69. **Assertion : Let $f: R \to R$ be a function such that $f(x) = x^3 + x^2 + 3x + \sin x$.**

Then f is one-one.

Reason : $f(x)$ neither increasing nor decreasing function.

70. **Consider the irrational numbers : π^e, e^π**

Assertion : $\pi^e < e^\pi$

Reason : $f(x) = x^{1/x}$ is decreasing in (e, ∞).

71. **Consider the equation $x^3 - 3x + k = 0$, $k \in R$.**

Assertion: There is no value of k for which the given equation has two distinct roots in (0, 1)

Reason : Between two consecutive roots of $f'(x) = 0$, (f(x) is a polynomial), f(x) = 0 must have one root.

72. **Assertion. : $f(x) = \frac{1}{x+1} - \ell n(1+x)$, $x > 0$ is a decreasing function.**

Reason. : $f'(x) < 0 \ \forall x > 0$

COMPREHENSION QUESTIONS

Consider the function

f : $(-\infty, \infty) \to (-\infty, \infty)$ defined by

$f(x) = \frac{x^2 - ax + 1}{x^2 + ax + 1}$, $0 < a < 2$ **[IIT - 2008]**

73. **Which of the following is true?**

A) $(2+a)^2 f''(1) + (2-a)^2 f''(-1) = 0$

B) $(2-a)^2 f''(1) - (2+a)^2 f''(-1) = 0$

C) $f'(1) f'(-1) = (2-a)^2$

D) $f'(1) f'(-1) = -(2+a)^2$

74. **Which of the following is true?**

A) f(x) is decreasing on (–1, 1) and has a local minimum at x = 1

B) f(x) is increasing on (–1, 1) and has a local maximum at x = 1

C) f(x) is increasing on (–1, 1) but has neither a local maximm nor a local minimum at x = 1

D) f(x) is decreasing on (–1, 1) but has neither a local maximum nor a local minimum at x = 1

75. **Let** $g(x) = \int_0^{e^x} \frac{f'(t)}{1+t^2} dt$

Which of the following is true?

A) $g'(x)$ is positive on $(-\infty, 0)$ and negative on $(0, \infty)$

B) $g'(x)$ is negative on $(-\infty, 0)$ and positive on $(0, \infty)$

C) $g'(x)$ changes sign on both $(-\infty, 0)$ and $(0, \infty)$

D) $g'(x)$ does not change sign on $(-\infty, \infty)$

Q.76 to Q.78

Suppose you do not know the function f (x), however some information about f(x) is listed below. Read the following carefully before attempting the questions

(i) f(x) is continuous and defined for all real numbers

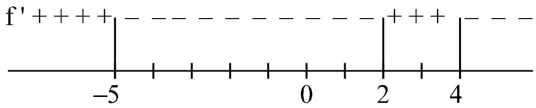
(ii) $f'(-5) = 0$; $f'(2)$ is not defined and

$f'(4) = 0$

(iii) (–5, 12) is a point which lies on the graph of f (x)

(iv) $f''(2)$ is undefined, but $f''(x)$ is negative everywhere else.

(v) the signs of $f'(x)$ is given below



76. **On the possible graph of $y = f(x)$ we have**

A) $x = -5$ is a point of relative minima.

B) $x = 2$ is a point of relative maxima.

C) $x = 4$ is a point of relative minima.

D) graph of $y = f(x)$ must have a geometrical sharp corner.

77. **From the possible graph of $y = f(x)$, we can say that**

A) There is exactly one point of inflection on the curve.

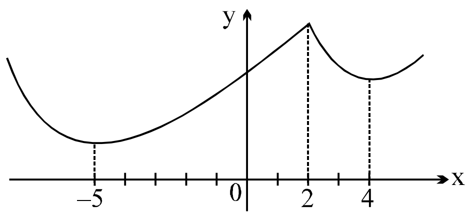
B) $f(x)$ increases on $-5 < x < 2$ and $x > 4$ and decreases on $-\infty < x < -5$ and $2 < x < 4$.

C) The curve is always concave down.

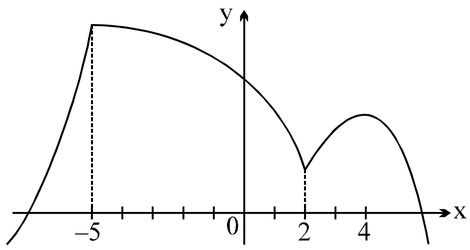
D) Curve always concave up.

78. **Possible graph of $y = f(x)$ is**

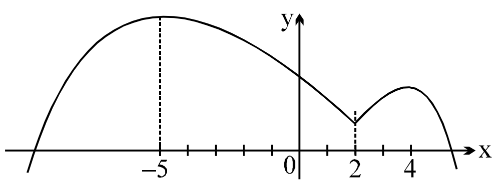
A)



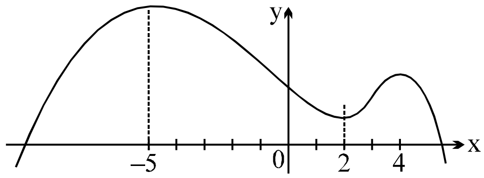
B)



C)



D)



INTEGER QUESTIONS

79. **If the chord joining the points where $x = p$, $x = -p$ on the curve $y = ax^2 + bx + c$ is parallel to the tangent drawn to the curve at (α, β) then α is**

80. **Area of the triangle formed by the tangent, normal at (1,1) on the curve $\sqrt{x} + \sqrt{y} = 2$ and the y - axis is**

81. **Abscissa of a point on the curve $9y^2 = x^3$ where normal to the curve makes equal intercepts with the axes is**

82. **If the curves $ax^2 + by^2 = 1$ and $a_1x^2 + b_1y^2 = 1$ may cut each other orthogonally such that $\frac{1}{a} - \frac{1}{a_1} = \lambda\left(\frac{1}{b} - \frac{1}{b_1}\right)$ then λ is equal to**

83. **The number of non - zero integral values 'a' for which the function $f(x) = x^4 + ax^3 + \frac{3x^2}{2} + 1$ is concave upward along the entire real line is**

84. **Let C be a curve defined by $y = e^{a+bx^2}$. The curve C passes trhoug the point $P(1,1)$ and the slope of the curve tan gent at P is -2. Then the value of $2a - 3b$ is**

85. **At a point $p(a, a^n)$ on the graph of $y = x^n$ in the first quadrant a normal is drawn. The normal intersets the line y-axis at the point $(0, b)$. If $\lim_{a \to 0} b = \frac{1}{2}$**

then $n =$

SUBJECTIVE QUESTIONS

86. Find the shortest distance of the point $(0,c)$ from the parabola $y = x^2$, where $0 \le c \le 5$.

87. If $ax^2 + \frac{b}{x} \ge c$ for all positive 'x' where a>0 and b>0 show that $27ab^2 \ge 4c^3$.

88. Find the maxima and minima of the function $y = x(x-1)^2, 0 \le x \le 2$.

89. You have been asked to design a one litre can shaped like a right circular cylinder what dimensions will use the least material.

90. Assuming that the stiffness of a beam of rectangular cross-section varies as the breadth and as the cube of the depth, prove that for the stiffest beam, breadth must be equal to half of the diameter of the log

91. Point on the curve $a^2x^2 + 4^2y^2 = 4a^2$ $\left(4 < a^2 < 8\right)$ that is farthest from point $(-2,0)$

92. $g(x) = 2f\left(x^2/2\right) + f\left(6 - x^2\right)$ $f''(x) > 0, \forall x \in R$ then nature of $g(x)$

93. Find the equations of vertical and inclined assymptotes of the curve $y = \frac{3x}{x-1} + 3x$

94. Find the extrema of $f(x) = \frac{50}{3x^4 + 8x^3 - 18x^2 + 60}$

95. Suppose it costs $c(x) = x^3 - 6x^2 + 15x$ dollars to produce x radiators when 8 to 30 radiators perday on that of revenue of $r(x) = x^3 - 3x^2 + 12x$ for selling x radiators how much extra will it cost produce for radiator that day profit (or) gain

96. A hot air ballon rising straight up from a level field is tracked by a range 500 ft from lift point at the moment range finder elevation angle is $\frac{\pi}{4}$ the angle is increasing at a rate of 0.14 rad/min. How fast ballon is rising at that moment.

97. A company accepts only rectangular boxes whole sum of length and width does not excude 108 inch. Find dimensions of box of largest volume

LEVEL - V

CLASS WORK

KEY

SINGLE ANSWER QUESTIONS

1) D	2) B	3)D	4) A
5)B	6)A	7) C	8) C
9) B	10) A	11) B	12) D
13) D	14) C	15) C	16) B
17) B	18) B	19) C	20) B
21)D	22) D	23)B	24)C
25)B	26)C	27) A	28) B
29) B	30) D	31)C	32)D
33) C	34) D	35)D	36) A
37) A	38) D	39) B	40) A
41) A	42) B	43) B	44) C
45) D			

MULTI ANSWER QUESTIONS

46) B,C	47) B
48) C ,D	49)A,C
50) A ,D	51) A,B,C
52) A,B,C	53) A,C
54) A,D	55) A,D
56) A , B, D	57) A,B,C
58) A,C	59) A ,D
60)B,D	61) A, B
62) A, B, C	

MATRIX- MATCHING QUESTIONS

63) $A \to P, B \to Q, C \to S, D \to R$

64) A-S,T , B-P, T, C-Q, T, D-R, T

65) A- Q, B- P, C- S, T ,D- R, T

66) A- P,Q,R,S,T , B- S , C- Q,R,P , D- P

67) A- T, B- T, C- P, D- Q

68) A- S, B- Q, T, C- P, T, D- R, T

ASSERTION & REASON QUESTIONS

69) C 70) A 71) C 72) A

COMPREHENSION QUESTIONS

73) A 74) A 75) B

76) D 77) C 78) C

INTEGER QUESTION

79) 0 80) 1 81) 4 82) 1

83) 3 84) 5 85) 2

SUBJECTIVE TYPE QUESTION

86) $D_{\min} = \sqrt{c - \frac{1}{4}}$

88) $y_{\max} = \frac{4}{27},\ y_{\min} = 0$

LEVEL - V

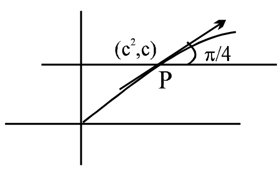
CLASS WORK

HINTS

SINGLE ANSWER QUESTIONS

1. $y = \sqrt{x}$ and $y = c$; solving $x = c^2$

$$y = \sqrt{x}\ ;\ \frac{dy}{dx} = \frac{1}{2\sqrt{x}}$$



$$\left.\frac{dy}{dx}\right]_P = \frac{1}{2c} = 1 \Rightarrow c = \frac{1}{2}$$

Equation of the line is $y = \frac{1}{2}$

2. $\frac{dx}{dt} = -\frac{2t}{(1+t)^2}$

$$\frac{dy}{dt} = -\frac{1+3t^2}{t^2(1+t^2)^2}$$

$$\frac{dy}{dx} = \frac{1+3t^2}{2t^3} > 0 \quad \text{for } t > 0$$

$\therefore$ y is increasing for every $x \in (0,1)$

3. $\frac{dy}{dx} = f'(x)_{(3,4)} \Rightarrow f'(3) = 1$

4. $f(0) > f(0+h)$

$f(0) > f(0-h)$

hence it is local maximum.

5. $e^x = 1 + \frac{x}{1} + \frac{x^2}{2!} + ... > 1 + x$

$$\log(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} < x$$

$$\sin x = x - \frac{x^3}{3!} + ... < x$$

6. $f'(x) = e^{x(1+x)} + (1-2x)xe^{x(1-x)}$

$\therefore$ f(x) is increasing on $\left[-\frac{1}{2}, 1\right]$

7. $f'(x) = 2e^{x^2-4x}(2x-4) < 0$ for $x < 2$

8. $f'(x) < 0 \ \forall\ x \in R$

$\Rightarrow (a+2)x^2 - 2ax + 3 < 0 \quad \forall\ x \in R$

$\Rightarrow$ discriminant < 0 and $a + 2 < 0$

9. Check the values at critical points $\left(\frac{dy}{dx} = 0\right)$ and extreme positions $\{0, 2\pi\}$

10. $f'(x) = 2x\log x + x > 0\ \forall x \in [1, e]$

11. Slope of normal $= -\frac{1}{\left(\frac{dy}{dx}\right)} = -\frac{a}{c}$

12. For L.M.V.T f(x) must be continuous [a, b] and differentiable in (a, b)

13. Function is not derivable at $x = 0$.

14. $$f(x) = \frac{-\cos 2\left(x + \frac{\pi}{4}\right)}{\sin\left(x + \frac{\pi}{4}\right)} = \frac{2\sin^2\left(x + \frac{\pi}{4}\right) - 1}{\sin\left(x + \frac{\pi}{4}\right)}$$

Let $\sin\left(x+\frac{\pi}{4}\right)=t,\ t\in\left[\frac{1}{\sqrt{2}},1\right]$

$f(t)=\frac{2t^2-1}{t}\Rightarrow f'(t)=2+\frac{1}{t^2}>0$

$\therefore$ maximum of f(t) occurs at t=1 and its value is 1.

15. $y=\cos^{-1}x\left(\frac{\pi}{2}-\cos^{-1}x\right)=t\left(\frac{\pi}{2}-t\right)$; $0\le t\le\pi$

16. $f(1)=f(2) \Rightarrow 1+b+c=8+4b+2c$

$f'(4/3)=0 \Rightarrow 3.\frac{16}{9}+2b\frac{4}{3}+c=0$

Solving both, we get b = -5, c = 8.

17. $f'(x)=ax^3+bx^2+cx+d$

$f(x)=\frac{ax^4}{4}+\frac{bx^3}{3}+\frac{cx^2}{2}+dx+e$

given $6a+4b+3c+3d=0$,

$f(+2)=e=f(0)$ use rolle's theorem.

18. $\frac{dy}{dx}=0$ for $x=-1$ and $x=2$

19. Let $P(t, t^4)$ be a point on the curve equation of tangnet at P, $y-t^4=4t^3(x-t)$

π it is drawn from (2, 0), then $t=0,\ \frac{8}{3}$

So, equation of required tangnet is

$y-\frac{4096}{81}=\frac{2048}{27}\left(x-\frac{8}{3}\right)$

20. A.M $\ge$ G.M; or By using maxima and minima concept

21. For $y^2=4ax$, y-axis is tangent at (0, 0), while for $x^2=4ay$, x-axis is tangent at (0, 0).

Thus the two curves cut each other at right angles.

22. $f'(x)=-x(x+2)e^{-x}=-(x+1)e^{-x}=0$

$\Rightarrow \quad x=-1$

For $x\in(-\infty,-1)$, $f'(x)>0$ and for

$x\in(-\infty,-1)$, $f'(x)<0$

$\therefore$ f(x) is increasing on $(-\infty,-1)$ and decreasing on $(-1,\ \infty)$

23. $\frac{dy}{dx}=25x^{24}(1-x)^{75}-75x^{25}(1-x)^{74}$

$=25x^{24}(1-x)^{74}(1-x-3x)$

$=25x^{24}(1-x)^{74}(1-4x)=\frac{dy}{dx}=0$

$x=\frac{1}{4}\in(0,1)$

$\therefore$ Max. value of y occurs at $x=\frac{1}{4}$

24. $f'(x)=\frac{\sin x-x\cos x}{\sin^2 x}$

Where $\sin^2x$ is always +ve, when $0<x\le1$.

But to check Nr., we again

let $h(x)=\sin x-x\cos x$.

$\Rightarrow\ h'(x)=x\sin x>0$ for $0<x\le1$

$\Rightarrow$ h(x) is increasing

$\Rightarrow$ $h(0)<h(x)$, when $0<x\le1$ $\Rightarrow 0<$ $\sin x-x\cos x$, when $0<x\le1$

$\Rightarrow$ f(x) is increasing on (0, 1].

25. $\Rightarrow\ f'(x)=4\sin^3x\cos x-4\cos^3x\sin x$

$=-\sin 4x>0\Rightarrow \quad \frac{\pi}{4}<x<\frac{\pi}{2}$

If f(x) increasing on $\left(\frac{\pi}{4},\frac{\pi}{2}\right)$ and $\left(\frac{\pi}{4},\frac{3\pi}{8}\right)$

26. $f'(x)<0\Rightarrow (x-1)(x-2)<0 \Rightarrow 1<x<2$

27. Let $f'(x)=ax^2+bx+c$

$\Rightarrow f(x)=\frac{2ax^3+3bx^2+6cx+d}{6}$

$\Rightarrow f(0)=f(1)=\frac{d}{6}$

28. $f(x)\ge 2x-4$, using Lagrange mean value theorem.

29. P(x) will cut x-axis k times, hence will give at least (k–1) maxima and minima.

30. Using the graph of $y=ax^2+bx+c$

31. tangent to $y=x^2+bx-b$ at (1, 1) is

$x-\text{int ercept}=\frac{b+1}{b+2}$

and y-intercept $= -(b+1)$

ATQ $Ar(\Delta) = 2$

$$\Rightarrow \frac{1}{2}\left(\frac{b+1}{b+2}\right)[-(b+1)] = 2$$

$$= b - 3$$

32. $m(b) = \dfrac{1}{1+b^2}$, Thus $m(b) = (0, 1]$

33. $x^2y = c^3$

$$x^2\frac{dy}{dx} + 2xy = 0 \Rightarrow \frac{dy}{dx} = -\frac{2y}{x}$$

equation of tangent at (x,y)

$$Y - y = -\frac{2y}{x}(X - x)$$

$$Y = 0,\ \text{gives},\ X = \frac{3x}{2} = a$$

and $\quad X = 0$, gives, $Y = 3y = b$

$$\text{Now } a^2b = \frac{9x^2}{4}.3y = \frac{27}{4}x^2y = \frac{27}{4}c^3 \Rightarrow \text{(C)}$$

34. $f(0) = -1$; $f(1) = 7$. So $f(0)$ and $f(1)$ have opposite sign $\Rightarrow$ (D)

35. $f(x)$ vanishes at points where $\sin\dfrac{\pi}{x} = 0$

i.e. $\dfrac{\pi}{x} = k\pi$, $k = 1, 2, 3, 4, \ldots..$

hence $x = \dfrac{1}{k}$. Also $f'(x) = \sin\dfrac{\pi}{x} - \dfrac{\pi}{x}\cos\dfrac{\pi}{x}$

if $x \neq 0$

Since the function has a derivative at any interior point of the interval (0, 1), also continuous in [0,1] and $f(0) = f(1)$ hence Rolle's theorem is applicable to any one of the interval $\left[\dfrac{1}{2}, 1\right]$,

$$\left[\frac{1}{3}, \frac{1}{2}\right], \ldots \left[\frac{1}{k+1}, \frac{1}{k}\right]$$

hence $\exists$ some c in each of these interval where $f'(c) = 0 \Rightarrow$ infinite points $\Rightarrow$ (D)

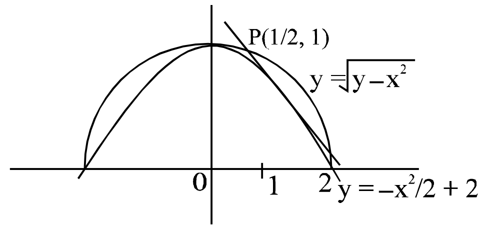
36. Using LMVT in [0, 2]

$$\frac{f(2) - f(0)}{2 - 0} = f'(c) \quad \text{where } c \in (0, 2)$$

$$\frac{f(2)+3}{2} \leq 5,\quad f(2) \leq 7 \Rightarrow \text{(A)}$$

37. Equation of line through (1/2, 2) is

$$y - 2 = m\left(x - \frac{1}{2}\right) \quad \ldots.(1)$$



Solve it with $y = -\dfrac{x^2}{2} + 2$

$$2 + m\left(x - \frac{1}{2}\right) = -\frac{x^2}{2} + 2$$

$$\frac{x^2}{2} + mx - \frac{m}{2} = 0$$

$D = 0$ gives

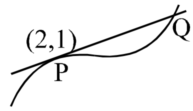
$$m^2 - 4\left(\frac{1}{2}\right)\left(-\frac{m}{2}\right) = 0 \Rightarrow m^2 + m = 0$$

$\Rightarrow \quad m = 0$ or $m = -1$

with $m = 0$, Tangent is $y = 2$ (rejected)

with $m = 1$, Tangent is $y = -x + \dfrac{5}{2}$

38. eliminating t gives $y^2(x-1) = 1$.



Equation of tangent at $P(2, 1)$ is $x + 2y = 4$.

Solving with curve $x = 5$ & $y = -1/2$

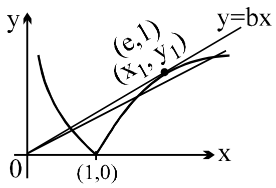
$$\Rightarrow Q\ (5, 1/2) \Rightarrow PQ = \frac{3\sqrt{5}}{2}$$

39. $\dfrac{dy}{dx} = 12x(x^2 - x + 1) + a \quad$ &

$$\frac{d^2y}{dx^2} = 12(3x^2 - 2x + 1) > 0$$

$\Rightarrow \dfrac{dy}{dx}$ is an increasing function. But $\dfrac{dy}{dx}$ is a polynomial of degree 3 $\Rightarrow$ exactly one root

40.

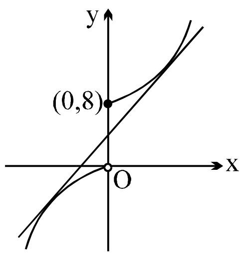


$1/x_1 = p = y_1/x_1 \Rightarrow y_1 = 1$ & $x_1 = e$.
Hence slope of $y = px$ i.e. $p \in (0, 1/e)$ for three roots

41. $\left|\dfrac{dy}{dx}\right| > 1$

42. $v = a^3 \Rightarrow \dfrac{dv}{dt} = 3a^2\dfrac{da}{dt}$

43.



Let $y = mx + c$ be a tangent to $f(x)$
$y = x^2 + 8$ for $x \geq 0$
$mx + c = x^2 + 8$
$x^2 - mx + 8 - c = 0$ (for the line to be tangent $D = 0$)
$\therefore \quad m^2 = 4(8 - c) \quad$ ....(1)
again $\quad y = -x^2$ for $x < 0$
$mx + c = -x^2$
$x^2 + mx + c = 0$
$D = 0 \Rightarrow \quad m^2 = 4c \quad$ ....(2)
from (1) and (2)
$c = 4, m = 4 \quad \therefore \quad y = 4x + 4$
put $y = 0 \quad \Rightarrow \quad x = -1$

44. Let x tree be added then
$P(x) = (x + 50)(800 - 10x)$
now $\quad P'(x) = 0 \Rightarrow \quad x = 15$

45. $y' = +\dfrac{1 \cdot 2\sin x\cos x}{(2+\cos^2 x)^2} = 0$

$\sin 2x = 0$

$x = \dfrac{n\pi}{2}, \; n \in I$

if n is odd then $y = \dfrac{1}{2+\cos^2 x} = \dfrac{1}{2}$

if n is even then $y = \dfrac{1}{2+\cos^2 x} = \dfrac{1}{3}$

MULTI ANSWER QUESTIONS

46. Differentiating w.r.t. x, $y + x\dfrac{dy}{dx} = 0$

$\therefore$ the equation of the normal at (α, β) is

$y - \beta = \dfrac{\alpha}{\beta}(x - \alpha)$ or $\alpha x - \beta y = \alpha^2 - \beta^2$

The given line is a normal at (α, β) if

$\dfrac{\alpha}{a} = -\dfrac{\beta}{b} = \dfrac{\alpha^2 - \beta}{-c}$

$\Rightarrow \dfrac{\alpha}{a} = \dfrac{\beta}{-b} = \dfrac{\sqrt{\alpha\beta}}{\sqrt{-ab}} = \dfrac{1}{\sqrt{-ab}} \quad (\because \alpha\beta = 1)$

$\therefore$ a, b are real if $ab < 0$ i.e., $a > 0, b < 0$ or $a < 0, b > 0$.

47. $f'(x) = 6\sin^2 x\cos x - 6\sin x\cos x + 12\cos x$
$= 6\cos x\{\sin^2 x - \sin x + 2\} =$

$6\cos x\left\{\left(\sin x - \dfrac{1}{2}\right)^2 + \dfrac{7}{4}\right\}$

$\therefore$ in $\left[0, \dfrac{\pi}{2}\right]$, $f'(x) \geq 0$. So, f(X) is increasing in $\left[0, \dfrac{\pi}{2}\right]$

48. Here, $f(x) = x^2 + 1, \; 1 \leq x < 2$

$\dfrac{x^2+1}{2}, 2 \leq x < 3$

$\dfrac{x^2+1}{3}, 3 \leq x \leq 3.9$

$f'(x) > 0$ in each of the intervals and so f(x) is increasing in each of the intervals.

$\therefore \; 2 \leq f(x) < 5$ in $1 \leq x < 2$; $\dfrac{5}{2} \leq f(x) < 5$ in $2 \leq x < 3$

$\frac{10}{3} \le f(x) \le \frac{1}{3} \times 16.21$ in $3 \le x \le 3.9$

Hence the least value is 2 and the greatest value is $\frac{1}{3} \times 16.21$

49. $h'(x) = 3f'(x)[\{f(x) - 1/3\}^2 + 2/9]$

Note that

$h'(x) < 0$ whenever $f'(x) < 0$ and $h'(x) > 0$ whenever $f'(x) > 0$, thus, h(x) increases (decreases) whenever f(x) increases (decreases).

50. (1, 0) is on both the curves.

So, $0 = 1 + a + b$ and $0 = c - 1$

For the first parabola, $\frac{dy}{dx} = 2x + a$

$\therefore \frac{dy}{dx})_{1,0} = 2 + a$

For the second parabola, $\frac{dy}{dx} = c - 2x$

$\therefore \frac{dy}{dx})_{1,0} = c - 2$

$\therefore 2 + a = c - 2$ and $0 = c - 1$

$\Rightarrow c = 1, a = -3 \therefore 0 = 1 + (-3) + b$ or $b = 2$

51. Differentiating w.r.t. x, $8x + 18y\frac{dy}{dx} = 0$

or $\frac{dy}{dx} = -\frac{4x}{9y}$

The tangent is equally inclined to the axes

$\Rightarrow \frac{dy}{dx} = \pm 1$

$\therefore \frac{4x}{9y} = \pm 1 \qquad \Rightarrow -\frac{4}{9}x, \frac{4}{9x}x$

$\therefore 4x^2 + 9.\frac{16}{81}x^2 = 36 \Rightarrow \frac{52}{9}x^2 = 36 \Rightarrow x = \pm\frac{9}{\sqrt{13}}$

52. $f'(x) = \begin{cases} 6x + 12 & ; \ -1 < x < 2 \\ -1 & ; \ 2 < x < 3 \end{cases}$

$f'(2^+) = -1, f'(2^-) = 24 \Rightarrow f'(2)$ do not exist.

$f'(x) > 0 \ \forall \ x \in (-1, 2)$

$LHL = 35, \ RHL = 35 \ \Rightarrow f(x)$ is cont. at $x = 2$

$f(2^+) < 0$ and $f(2^-) > 0 \Rightarrow$ maxima at $x = 2$

53. $f'(x) = 3x^2 + 2ax + b + 5\sin 2x$

f(x) increases always, so $f'(x) > 0 \ \forall \ x \in R$

$\Rightarrow 3x^2 + 2ax + b + 5\sin 2x > 0$

which will be true if $3x^2 + 2ax + b - 5 > 0$, always if $D < 0$

54. $f'(x) = 4(x-1)^3 . (x-2)^n . n(x-2)^{n-1}$

$= 4(x-1)^3 . (x-2)^{n-1} (4x - 8 + nx - n)$

$f'(x) = 0$ if $f = 1, 2, \frac{8+n}{4+n}$

$f'(x)(1-\in) = (-\in)^3 .(-1-\in)^{n-1} .\{(4+n)(1-\in) - 8 - n\}$

$= (-ve) . (-ve)^{n-1} .\{(4+n)\in - 4 - n\}$

$= (-ve)^{n-1}$

$f'(x)(1+\in)$

$= \in^3 . (-1+\in)^{n-1} .\{(4+n)(1+\in) - 8 - n\}$

$= (+ve)(-ve)^{n-1}.\{(4+n)\in - 4 - n\} = (-ve)^{n-1} . (-ve) = (-ve)^n$

therefore $f'(1-\in) > 0, f'(1+\in) < 0$ if n is odd.

So, at $x = 1$ there is a maximum if n is odd.

Similary we can prove there is a minimum at $x = 2$ when n is even.

55. $f'(x) = \begin{cases} 3x^2 + 2x^{-10} & ; -1 < x < 0 \\ \cos x & ; 0 < x < \frac{\pi}{2} \\ -\sin x & ; \frac{\pi}{2} < x < \pi \end{cases}$

f(x) is continuous everywhere decide sign of $f'(x)$ about 0 and $\frac{\pi}{2}$

−		+		−
dec	0	inc.	$\frac{\pi}{2}$	dec
	minima		maxima	

to find absolute maxima or minima compare f(-1), f(0), $f\left(\frac{\pi}{2}\right)$, f(p)

56. We know that sinx is an increasing funciton of

x in $\left[-\frac{\pi}{2},\frac{\pi}{2}\right]$

$f\{f(x)\} = \sin(\sin x)$;

$\therefore \frac{d}{dx}\{f(f(x)\} = \cos(\sin x) . \cos x \geq 0$ for

$-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$

$\therefore$ f{f(x)} is increasing $\left[-\frac{\pi}{2},\frac{\pi}{2}\right]$

57. Solution : a , b , c

$\Rightarrow f'(x) = 4x^3 - 4 = 4(x-1)(x^2+x+1)$.

so f (x) decreases in $(-\infty,1)$ and increases in $(1,\infty)$ But f (1) = -4 (which is minimum value of f (x)). So f (x) = 0 has only two real roots in which one will be positive and one will be negative (as f (0) is negative)

58. Solution : $f'(x) = 0$

$\Rightarrow (\sin x + \cos x)$ (non - quantity) = 0

$\Rightarrow \tan x = -1 \Rightarrow \quad x = \frac{3\pi}{4}$ or $\frac{7\pi}{4}$

Global minimum $= x = 2n\pi + \left(\frac{3\pi}{4}\right)$

Global maximum $= x = 2n\pi + \left(\frac{7\pi}{4}\right)$

$M = \frac{4}{8-\sqrt{2}}, m = \frac{4}{8+\sqrt{2}}$

Hence, (a) and (c) are the correct answers.

59. $f''(x) < 0 \quad \forall x \in R$.

$\Rightarrow \ f'(x)$ is decreasing function and

$g'(x) = 2x\left(f'(x^2-2) - f'(6-x^2)\right)$

for $g'(x) > 0$ (i) $x > 0$ and $x^2 - 2 < 6 - x^2$ or (ii) $x < 0$ and $x^2 - 2 > 6 - x^2$

60. $\frac{dy}{dx} = f'(x) \Rightarrow x(e^x-1)(x-1)(x-2)^3(x-3)^5 = 0$

Critical points are 0, 1, 2, 3. Consider change of sign of $\frac{dy}{dx}$ at $x = 3$

$x < 3, \frac{dy}{dx} = -ve$ and $x > 3, \frac{dy}{dx} = +ve$

Change is from –ve to +ve,
Hence minimum at x = 3.
Again minimum and maximum occur alternately.
$\therefore$ 2nd minimum is at x = 1.

61. Conceptual

62. $f(x) = \frac{\sin(x+a)}{\sin(x+b)}$

$f'(x) =$

$$\frac{\sin(x+b)\times\cos(x+a) - \sin(x+a)\cos(x+b)}{\sin^2(x+b)}$$

$= \frac{\sin(b-a)}{\sin^2(x+b)}$

If $\sin(b-a) = 0$ then $f'(x) = 0 \Rightarrow f(x)$ will be constant

i.e. $b - a = n\pi$ or $n\pi$ or $b - a = (2n+1)\pi$ or $b - a = 2n\pi$

then $f(x)$ has no minima

MATRIX- MATCH QUESTIONS

63. A) $f''(x) > 0 \forall x \in R \Rightarrow f'(x)$ is an increasing function.

Now $g'(x) = -f'(4-x) + f'(2+x)$

If $g'(x) > 0 \Rightarrow f'(2+x) > f'(4-x) \Rightarrow 2+x > 4-x$ or $x > 1$

B) $f'(x) = 3(x-1)(x+1) \Rightarrow f'(x) = 0$

has roots x=-1, 1

f(x) =0 will have exactly one real root if f(-1) f(1) >0

$\Rightarrow (a+2)(a-2) > 0 \Rightarrow a < -2$ or $a > 2$

C) $f'(x) = -\sin x + a^2 \geq 0 \forall x \in R$

$\Rightarrow a^2 \geq \sin x \forall x \in R$

D) $f'(x)=2e^{x}+ae^{-x}+2a+1=2e^{-x}(e^{x}+a)\left(e^{x}+\frac{1}{2}\right)$

f(x) increases for all x if $f'(x)\geq 0 \forall x\in R$

$\therefore e^{x}+a\geq 0\ \forall\ x\in R\Rightarrow a\geq 0$

64. $f'(c)=\frac{f(b)-f(a)}{b-a}$ for LMVT

$f'(c)=\frac{0+1}{3}$ for 'a'

$\Rightarrow 2c-2=\frac{1}{3}\Rightarrow 2c=\frac{7}{3}\Rightarrow c=\frac{7}{6}$

$f'(c)=\frac{1}{3}$ for 'b'

$\Rightarrow 2-2c=\frac{1}{3}\Rightarrow 2c=\frac{5}{3}\Rightarrow c=\frac{5}{6}$

$f'(c)=\frac{1-3}{+1}\Rightarrow 3c^{2}-4c-1=-2$

$3c^{2}-4c+1=0\Rightarrow c=1$

$3c(c-1)-1(c-1)=0$

$c=\frac{1}{3}\qquad \because c\in[0,1]$

$f'(c)=\frac{6}{3}\Rightarrow 3c^{2}-12c+11=2$

$\Rightarrow c^{2}-4c+3=0,\quad c=3\ \because c\in[1,4]$

65. A. $\frac{(x-1)(x-2)}{(x-3)}=f(x)$

$f'(x)=\frac{(x-3)[x-2+x-1]-(x-1)(x-2)}{(x-3)^{2}}$

$=\frac{(x-3)(2x-3)-(x-1)(x-2)}{(x-3)^{2}}$

$=\frac{x^{2}-6x+7}{(x-3)^{2}}\Rightarrow x^{2}-6x+7<0$

when $x\in\left(3-\sqrt{2},3+\sqrt{2}\right)$

$\therefore$ Decreases when $x\in\left(3-\sqrt{2},3+\sqrt{2}\right)$

B. $\frac{(x-2)(x-3)}{(x-1)}$

$f'(x)=\frac{(x-1)(2x-5)-(x-2)(x-3)}{(x-1)^{2}}$

$=\frac{x^{2}-2x-1}{(x-1)^{2}}\qquad x=\frac{2\pm\sqrt{4+4}}{2}=1\pm\sqrt{2}$

$\Rightarrow$ Decreases when $x\in\left(1-\sqrt{2},1+\sqrt{2}\right)$

C. $\frac{(x-1)(x-3)}{(x-2)}$

$f'(x)=\frac{(x-2)(2x-4)-(x-1)(x-3)}{(x-2)^{2}}$

$=\frac{x^{2}-4x+4}{(x-2)^{2}}=1$

$\therefore$ Increases when $x\in R$, monotonic function

D. $\frac{(x-2)}{(x-1)(x-3)}$

$f'(x)=\frac{(x-1)(x-3)-(x-2)(2x-4)}{(x-2)^{2}(x-3)^{2}}$

$=\frac{-\left(x^{2}-4x+4\right)^{2}}{(x-1)^{2}(x-3)}<0$

$\therefore$ Decreases $\forall x\in R$, monotnic function

66. A. $f(x)=\sin x+\cos x+2x$

$f'(x)=\cos x-\sin x+2>0$

$\therefore$ Strictly increases on $(-\infty,\infty)$

B. $f(x)=\frac{x^{2}+x+1}{x^{2}-x+1}$

$$f'(x)=\frac{(2x+1)\left(x^2-x+1\right)-\left(x^2+x+1\right)(2x-1)}{\left(x^2-x+1\right)^2}$$

$$=\frac{-4x^2+2x^2+2}{\left(x^2-x+1\right)^2}=\frac{-2x^2+2}{\left(x^2-x+1\right)^2}>0$$

$$\Rightarrow x\in(-1,1)$$

C. $f(x)=\dfrac{x^2+x+1}{x^2-x+1}$ stricIty

decreases on

$(-\infty,-1)\cup(1,\infty)$

D. $f(x)=\dfrac{x^3}{x^4+27}$

$$f'(x)=\frac{\left(x^4+27\right)\left(3x^2\right)-x^3\left(4x^3\right)}{\left(x^4+27\right)^2}$$

$$=\frac{81x^2-x^6}{\left(x^4+27\right)^2}\Rightarrow x^2\left(81-x^4\right)>0\Rightarrow |x|<3$$

$$\Rightarrow x\in(3,\infty)\cup(-\infty,-3)$$

67. A. $f(x)=\dfrac{x}{1+x\tan x}$

$$f'(x)=\frac{(1+\tan x)-x\left((\tan x)+\sec^2 x\right)}{(1+x\tan x)^2}=0$$

$1-x^2\sec^2 x=1$

$x^2\sec^2 x=1$

$|\sec x|=\dfrac{1}{x}\ \Rightarrow x=|\cos x|$

$\therefore$ x-coordinate of poitn of intersection of $y=x, y=\cos x$

B. $f(x)=\left(\dfrac{x}{1+x\tan x}\right)^{-1}$

$$=\frac{1+x\tan x}{x}$$

$$f'(x)=-{}^{1}\!/_{x^2}+\sec^2 x=0\Rightarrow \sec x=\frac{1}{|x|}$$

$\Rightarrow$ x-coordinate of point of intersection of $y=x, y=\cos x$

C. $f(x)=\dfrac{x}{1+x\cot x}=\dfrac{x\tan x}{\tan x+x}$

$$f'(x)=\frac{(1+\tan x)\left(\tan x+x\sec^2 x\right)-x\tan x\left(\sec^2 x+1\right)}{(\tan x+x)^2}=0$$

$$\Rightarrow \frac{\tan^2 x+x^2\sec^2 x}{(\tan x+x)^2}>0$$

$\therefore$ Only real value in the domain of definition of the function.

D. $f(x)=\left(\dfrac{x}{1+x\cot x}\right)^{-1}$

$$f(x)=\frac{1+x\cot x}{x}=\frac{\tan x+x}{x\tan x}$$

$$f'(x)=\frac{\left(\sec^2 x+1\right)x\tan x-(\tan x+x)\left(\tan x+x\sec^2 x\right)}{x^2\tan^2 x}$$

$$=-\frac{\tan^2 x-x^2\sec^2 x}{x^2\tan^2 x}<0$$

$\therefore$ decreasing

$\therefore$ minimum when $x=\dfrac{\pi}{4}$

68. A. $f(x)=\dfrac{x}{4+x+x^2}$ on $[0,\infty)$

$$f'(x)=\frac{\left(4+x+x^2\right)-x(2x+1)}{\left(4+x+x^2\right)^2}=0$$

$x^2+x+4-2x^2-x=0$

$4=x^2, x=\pm 2$

if $x=2\Rightarrow \dfrac{2}{4+2+4}=\dfrac{2}{10}={}^{1}\!/_{5}$

$x=-2 \Rightarrow \frac{-2}{6}=-\frac{1}{3}$

B. $f'(x)=\frac{1-\ln x}{x^2}>0$

$\ln x<1 \qquad x<e$

$\therefore$ Maximum value of $\frac{\ln x}{x}=\frac{1}{e}$

C. $y=\frac{1}{x} \Rightarrow \frac{3}{e^3}x+\frac{27e}{x}$

$f'(x)=\frac{3}{e^3}-\frac{27e}{x^2}=0 \quad , \quad x\neq 0$

$x=3e^2$

$f''(x)=+\frac{54e}{x^3} \Rightarrow \frac{3}{e^3}\times 3e^2+27\times e\times\frac{1}{3e^2}$

$=\frac{9}{e}+\frac{9}{e}=\frac{18}{e}$

D. $R\theta+2R=4e$

Area $=\frac{1}{2}LR.\frac{1}{2}R^2\theta$

$\theta=\frac{4R-2R}{R} \Rightarrow \frac{1}{2}\left(4eR-2R^2\right)$

$=2eR-R^2=f(R)$

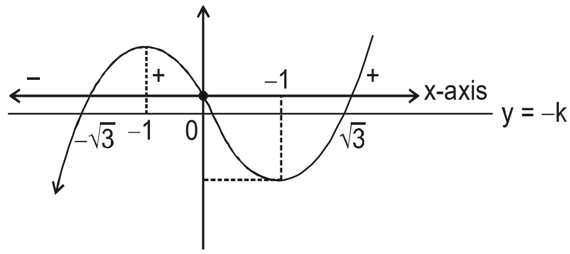
$f'(R)=2e-2R=0 \qquad R=e$

ASSERTION & REASON QUESTIONS

69. $f'(x)=2(x^2+1)+(x+1)^2+\cos x>0$ $\forall x\in R$.

70. $f'(x)<0$ for $x>e$

71. say $f(x)=x^3-3x=x(x-\sqrt{3})(x+\sqrt{3})$
Now $f(x)=-k$
will have roots when $y=-k$ and $f(x)=y$ cuts
$f'(x)=0$ at $x=\pm 1$



If $0<x<1$
f(x) is decreasing so $y=-k$ will cut at only one place moreover, f(x) is one-one between two consecutive exrema.

72. $f'(x)=-\frac{1}{(x+1)^2}-\frac{1}{x+1}$ which is negative for positive x.

COMPREHENSION QUESTIONS

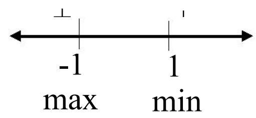
73. $f(x)=1-\frac{2ax}{x^2+ax+1} \Rightarrow f'(x)=\frac{2a(x^2-1)}{(x^2+ax+1)^2}$

$\Rightarrow f''(x)=$

$\frac{(x^2+ax+1)\left[(x^2+ax+1)4ax-4a(x^2-1)(2x+a)\right]}{(x^2+ax+1)^4}$

$f''(1)=4a/(2+a)^2$; $f''(-1)=-4a/(2-a)^2$

74. $f'(x)=\frac{2a(x^2-1)}{(x^2+ax+1)^2}$



So f(x) is decreasing in (-1, 1)
and local minima at x = 1

75. $g(x)=\int_0^{e^x}\frac{f'(t)}{1+t^2}$

$g'(x)=\frac{f'(e^x)}{1+(e^x)^2}e^x$

$f'(e^x)=\frac{2a(e^{2x}-1)}{(e^{2x}+ae^x+1)^2}$

76-78. At $x=-5$ f'(x) changes from + ve to – ve and $x=4$, f'(x) change sign for + ve to – ve hence maxima at $x=-5$ and 4. f is continuous and f'(x) is not defined hence $x=2$ must be geometrical sharp corner

INTEGER QUESTIONS

79. Points are

$\left(p, ap^2+bp+c\right), \left(-p, ap^2-bp+c\right)$

slope of the line joining the point $=\dfrac{2bp}{2p}=b$

$\dfrac{dy}{dx}=2ax+b$

$\left(\dfrac{dy}{dx}\right)_{(\alpha,\beta)}=2a\alpha+b$

$2a\alpha+b=b\Rightarrow\alpha=0$

80. Find equation of tangent and normal & then put x = 0 to evaluate vertices of triangle. Then find area of triangle.

81. Let (x_1,y_1) be a point on the curve

$9y_1^{\,2}=x_1^{\,3}$

$\left(\dfrac{dy}{dx}\right)_{(x_1,y_1)}=\dfrac{x_1^{\,2}}{6y_1}$ $\qquad \dfrac{1}{\left(\dfrac{dy}{dx}\right)_{(x_1,y_1)}}=\pm1$

$x_1=0,4$

but the line making equal intercepts with the axes can not pass through the origin $x_1=4$

82. Solve the curves simultaneously and apply

$\left.\dfrac{dy}{dx}\right|_{1st}\times\left.\dfrac{dy}{dx}\right|_{2nd}=-1$

83. Soluiton : **3**

$f''(x)=12x^2+6ax+3>0\,\forall x\in R$

$\Rightarrow 36a^2-144<0\Rightarrow a\in(-2,2)$

$\Rightarrow$ Number of non - zero integral values of 'a' is 3

84. $y=e^{a+bx^2}$

$1=e^{a+b}$ ($\because$ it passes through 1, 1)

$a+b=0$

$\left(\dfrac{dy}{dx}\right)_{(1,1)}=-2$

$e^{a+bx^2}.2bx=-2$

$e^{a+b}.2b(1)=-2$

$b=-1,a=1\quad 2a-3b=5$

85. $y=x^n$

$\dfrac{dy}{dx}=nx^{n-1}=na^{n-1}$

slope of normal $=-\dfrac{1}{na^{n-1}}$

equation of normal

$=y-a^n=-\dfrac{1}{na^{n-1}}(x-1)$

Put $x=0$ to get y-intercept

$y=a^n+\dfrac{1}{na^{n-2}}$

$\therefore b=a^n+\dfrac{1}{na^{n-2}}$

$$\lim_{a\to0}b=\begin{cases}0 & if\ \ n<2\\ \dfrac{1}{2} & if\ \ n=2\\ \infty & if\ \ n>2\end{cases}$$

SUBJECTIVE QUESTIONS

86. $(0,c)\ y=x^2,\ 0\le c\le5$

Any point on the parabols is (x,x^2)

Distance between (x,x^2) and $(0,c)$ is

$D=\sqrt{x^2+(x^2-c)^2}$

$\Rightarrow D^2=x^4-(2c-1)x^2+c^2=\left(x^2-\dfrac{2c-1}{2}\right)^2+c-\dfrac{1}{4}$

which is min. when $x^2-\dfrac{2c-1}{2}=0$

$\Rightarrow D_{\min}=\sqrt{c-\dfrac{1}{4}}$

87. given $ax^2+\dfrac{b}{x}\ge c\ \forall x>0,\ \ a>0,\ b>0$ .. (1)

to show that $27ab^2\ge4c^3$

Let us consider the function

$f(x) = ax^2 + \frac{b}{x} - c$ then

$$f'(x) = 2ax - \frac{b}{x^2} = 0 \Rightarrow x^3 = \frac{b}{2a} \Rightarrow x = \left(\frac{b}{2a}\right)^{1/3}$$

Also

$$f''(x) = 2a + \frac{2b}{x^3} \Rightarrow f''\left(\left(\frac{b}{2a}\right)^{1/3}\right) = 2a + \frac{2b}{b} \times 2a = 6a > 0$$

$\therefore$ f is min at $x = \left(\frac{b}{2a}\right)^{1/3}$

As (1) is true $\forall x$ $\therefore$ is so for $x = \left(\frac{b}{2a}\right)^{1/3}$

$$\Rightarrow a\left(\frac{b}{2a}\right)^{2/3} + \frac{b}{(b/2a)^{1/3}} \geq c$$

$$\Rightarrow \frac{a\left(\frac{b}{2a}\right) + b}{(b/2a)^{1/3}} \geq c \Rightarrow \frac{3b}{2}\left(\frac{2a}{b}\right)^{1/3} \geq c$$

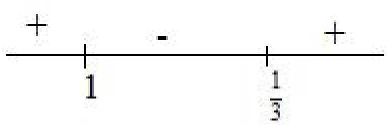
As a,b are +ve, cubing both sides, we get

$$\Rightarrow \frac{27b^3}{8}\frac{2a}{b} \geq c^3 \Rightarrow 27ab^2 \geq 4c^3$$

88. we have z $y = x(x-1)^2$, $0 \leq x \leq 2$

$$\frac{dy}{dx} = (x-1)^2 + 2x(x-1) = (x-1)(3x-1)$$

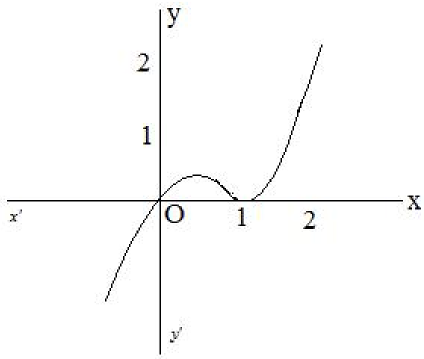
sign scheme of $f'(x)$ is



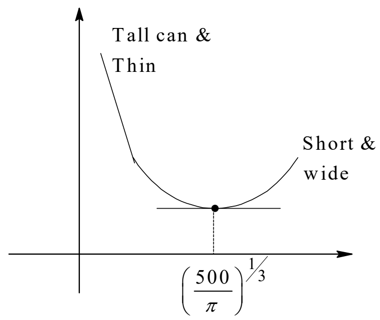
clearly, x=1 is the point of maxima (local) and x=1/3 is the point of minima (local).

$\therefore$ maximu value of y is $\frac{1}{3}\left(\frac{1}{3}-1\right)^2 = \frac{4}{27}$

minimum value of y is $1(1-1)^2 = 0$.



89.



Surface area $= 2\pi r^2 + 2\pi rh$

For a first approximation, we can ignore the thickness and waste of material.

$\therefore$ Volume $= \pi r^2 h = 100ml \Rightarrow h = \frac{1000}{\pi r^2}$

$$A = 2\pi r^2 + 2\pi rh$$

$$= 2\pi r^2 + 2\pi r\left(\frac{1000}{\pi r^2}\right) = 2\pi r^2 + \frac{200}{r}$$

$$A' = 4\pi r - \frac{2000}{r^2} = 0$$

$$2000 = 4\pi r^3$$

$$r = \left(\frac{500}{\pi}\right)^{1/3} \qquad \frac{d^2A}{dr^2} > 0$$

$$\Rightarrow h = \frac{1000}{\pi r^2} = 2\sqrt[3]{\frac{500}{\pi}} = 2r$$

$\therefore r \cong 5.42cm,\ h \cong 10.84cm$

90 Let the diameter be $2a$(of the log)

$\Rightarrow x^2 + y^2 = 4a^2$

$s \propto xy^3$

$s = \lambda xy^3$

$s = \lambda x\left(4a^2 - x^2\right)^{3/2}$

$s^2 = \lambda^2 x^2\left(4a^2 - x^2\right)^3$

$$\frac{ds^2}{dx} = \lambda^2\left[2x\left(4a^2 - x^2\right)^3 - 6x^3\left(4a^2 - x^2\right)^2\right]$$

$= 2x\lambda^2\left(4a^2 - x^2\right)^3\left(4a^2 - 4x^2\right)$

$= 8\lambda^2 x\left(4a^2 - x^2\right)^2\left(a^2 - x^2\right)$

$\frac{ds^2}{dx} = 0,\ \Rightarrow x = a\quad (\because x \ne 0, x \ne 2a)$

$\therefore$ Stiffest beam has breadth of beam as half the diameter.

91. $(2,0)$

$P(2\cos\theta, a\sin\theta)$ be a point on the curve

$Ap^2 = 4(1+\cos\theta)^2 + a^2\sin^2\theta$

$$\frac{d}{d\theta}\left(Ap^2\right) = 8(-\sin\theta)(1+\cos\theta) + a^2\sin^2\theta$$

$= \sin\theta\left(2a^2\cos\theta + 8(1+\cos\theta)\right) = 0$

$\Rightarrow \sin\theta = 0, \theta = 0$

$Ap^2 = 4(2)^2 + 0$, $(2,0)$ is the point.

$Ap = 4$

92. $g'(x) = 4xf'\left(x^2/2\right) - 2xf'\left(6 - x^2\right)$

$\Rightarrow f\left(x^2/2\right) > f'\left(6 - x^2\right)$

$\Rightarrow x^2/2 > 6 - x^2$ ($f'(x)$ is increasing)

$x^2 - 4 > 0 \quad (-\infty,-2)\cup(2,\infty)$

$f'\left(x^2/2\right) < f'\left(6 - x^2\right)$

$x^2 - 4 < 0 \quad \Rightarrow x \in (-2,2)$

$g(x)$ decreasing on $(-\infty,-2)\cup(0,2)$

increasing on $(-2,0)\cup(2,\infty)$

93. $\lim\limits_{x\to 1^-} y = \lim\limits_{x\to 1^-}\left(\frac{3x}{x-1} + 3x\right) = -\infty$

$\lim\limits_{x\to 1^+} y = \lim\limits_{x\to 1^+}\left(\frac{3x}{x-1} + 3x\right) = +\infty$

Curve has vertical asymptote at $x = 1$

Inclined asymptotes.

$\lim\limits_{x\to+\infty}\frac{f(x)}{x} = k_1$

$\lim\limits_{x\to+\infty} f(x) - k_1 x = b_1$

$y = k_1 x_1 + b_1$ is inclined asymptote

$k_1 = \lim\limits_{x\to\pm\infty}\frac{y}{x} = \lim\limits_{x\to\pm\infty}\frac{y}{x}\left(\frac{3}{x-1} + 3\right) = 3$

$b = \lim\limits_{x\to\pm\infty}(y - kx)$

$= \lim\limits_{x\to\pm\infty}\left(\frac{3x}{x-1} + 3x - 3x\right)$

$y = 3x + 3$ is inclined asymptote

94. let $f_1(x) = 3x^4 + 8x^3 - 18x^2 + 60$

$f_1'(x) = 12\left(x^2 + 2x - 3\right)$

$f_1''(x) = 12\left(3x^2 + 4x - 3\right)$

critical points $x_1=-3\ \ x_2=0\ \ x_3=1$

$f''(-3)<0\ \ f(x)$ has maximum

$f(-3)=\frac{-2}{3}\quad f(0)=5/6$

$f(1)=50/53$

95. The cost for producing one more radiator when already 10 radiators are produced is

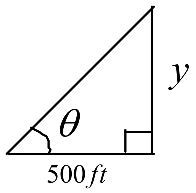
$c'(10)=3(10)^2-12(10)+15=195$

where as revenue will be

$r'(10)=3(100)-6(10)+12=252$

he gets a 57 dollors profit

96.



$\frac{y}{500}=\tan\theta$

$\tan\theta=y/500\qquad y=500\tan\theta$

$\frac{dy}{dt}=500\sec^2\theta\frac{d\theta}{dt}=500\times2\times0.14$

$=140\,ft/\min$

97. $x=$ length, $y=$ width, $z=$ height

$x+2y+2z=108$

$v(y,z)=(18-2y-2z)yz$

$=108yz-2y^2z-2yz^2$

$v_y(y,z)=108z-4yz-2z^2=0$

$v_z(y,z)=108y-2y^2-4yz=0$

$V_{yy}(y,z)=-4z\ \ V_{yz}=108-4y-4z$

$Y_{zz}=-4y$

where volume

$=36(18)(18)=11,664\,(in)^3$